

2.2.2 Algorithm vs Pseudocode

الگورتھم بتاتا ہے کہ کیا کرنا ہے، جبکہ سوڈو کوڈ (Pseudocode) بتاتا ہے کہ اسے کیسے کیا جا سکتا ہے۔

CS XI

Chapter 2: Computational Thinking & Algorithm

Lecture 02 of 11

★ Golden

2.2.2 Algorithm vs Pseudocode ★ Golden

LEARN IT

Both algorithms and pseudocode are used in problem-solving, but they serve different purposes in the design phase. The **algorithm** says *what* must be done. The **pseudocode** shows *how* it could be implemented, using programming-like keywords such as `IF`, `FOR` and `WHILE`, but it still cannot be run on a computer.

Point	Algorithm	Pseudocode
Definition	A step-by-step logical procedure to solve a problem.	A human-readable representation of an algorithm using plain English & keywords.
Nature	Conceptual and language-independent.	Semi-formal and structured, but not executable code.
Purpose	To define what needs to be done logically.	To present how it can be implemented in a readable format.
Form	Usually described in plain steps, bullets or numbered points.	Written using structured constructs (<code>IF</code> , <code>FOR</code> , <code>WHILE</code>) like programming.
Focus	Focuses on the logic and steps of the problem.	Focuses on representing steps clearly before coding.
Audience	For anyone trying to understand the logic (including non-programmers).	Mainly for programmers or students preparing for coding.
Format rules	No fixed format; can be informal.	Follows standard pseudocode conventions for clarity.

WORKED EXAMPLE

The same problem written both ways: print whether a number is even or odd.

Algorithm

Step 1: Start
Step 2: Read a number N
Step 3: Divide N by 2 and find the remainder
Step 4: If the remainder is 0, print "Even", otherwise print "Odd"
Step 5: End

Pseudocode

```
START
INPUT N
SET R TO N MOD 2
IF R = 0 THEN
    PRINT "Even"
ELSE
    PRINT "Odd"
END IF
END
```

The equivalent Python program (real, executable code):

```
n = 7
if n % 2 == 0:
    print("Even")
else:
    print("Odd")
```

OUTPUT

Odd

Problem → Algorithm (what) → Pseudocode (how) → Program (code) → Output

REMEMBER IT

Teacher's analogy: an **algorithm** is like a recipe ("Bake a cake at 350 degrees").

Pseudocode is like a structured kitchen manual using standard verbs (SET temperature TO 350; WHILE cake is not baked, WAIT). The program is the oven actually doing it.

COMMON MISTAKE

Pseudocode is **not** executable. If you type pseudocode into Python, it will give errors. Also, do not mix exact Python syntax (colons, brackets) into pseudocode answers unless asked — use keywords like IF ... THEN ... END IF.

اردو میں سمجھیں

الگورتھم بتاتا ہے کہ کیا کرنا ہے، جبکہ سوڈو کوڈ (Pseudocode) بتاتا ہے کہ اسے کیسے کیا جا سکتا ہے۔ الگورتھم سادہ مرحلوں میں لکھا جاتا ہے اور ہر کوئی اسے سمجھ سکتا ہے۔ سوڈو کوڈ میں IF، FOR اور WHILE جیسے الفاظ استعمال ہوتے ہیں اور یہ زیادہ تر پروگرامرز کے لیے ہوتا ہے، لیکن اسے کمپیوٹر پر چلایا نہیں جا سکتا۔

EXAM TIP

"Differentiate between algorithm and pseudocode" is a favourite 4–5 mark question. Draw a two-column table and give at least **five** points (definition, nature, purpose, form, audience). Adding the recipe vs kitchen-manual analogy and one small example of each earns full marks.

CHECK YOURSELF

1. Which one — algorithm or pseudocode — uses keywords like FOR and WHILE?
2. Which one is meant for non-programmers too?
3. Can pseudocode be executed by a computer?

Answers: 1. Pseudocode. 2. Algorithm. 3. No; it is semi-formal and not executable.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Pseudocode is:

- a. English-like steps for an algorithm, not a real language
- b. Runnable C++
- c. A logic gate
- d. A pie chart

MCQ 2. An algorithm vs pseudocode: algorithm is:

- a. The idea/method
- b. Always Python
- c. Always a table
- d. Always OSI

MCQ 3. Pseudocode should be:

- a. Clear enough to code from
- b. Encrypted
- c. Drawn as NAND only
- d. A CSV file

MCQ 4. Keywords like IF, WHILE in pseudocode are:

- a. Control, written in capitals by convention
- b. Python-only
- c. SQL-only
- d. Illegal

MCQ 5. You write pseudocode:

- a. Before coding
- b. After the program crashes only
- c. Instead of requirements
- d. On OSI layer 1

MCQ 6. A flowchart is:

- a. A diagram of the algorithm
- b. A database index
- c. A stack pop
- d. A sprint

Extra short questions

1. Differentiate algorithm and pseudocode.
2. Give two pseudocode conventions.
3. Why not write Python immediately?
4. Write 5 lines of pseudocode to add two numbers.

Extra long question

1. Write an algorithm AND pseudocode to find the largest of three numbers. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Algorithm: the method. Pseudocode: a structured English write-up of that method that a programmer can translate.
2. Capital keywords (IF, WHILE); one step per line; indent blocks; START/END.
3. Pseudocode is language-independent and easier to correct on paper.
4. START; READ a,b; SET $s \leftarrow a+b$; PRINT s; END.

Long

1. Algorithm numbered 1...n. Pseudocode with IF/ELSE. Same logic. Show a dry-run with 4, 9, 7 $\rightarrow$ 9.

Next lecture: 2.3.1 Algorithmic Strategy: Decomposition